

Comprehensive Customer Churn Analysis

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1. Data Integrity & Pre-processing

Before beginning the analysis, a rigorous data cleaning pipeline was implemented using Python (Pandas) to ensure 100% accuracy in findings.

- ✓ **Cleaning Done!** No duplicate records were found or retained.
- ✓ **Consistency Check:** Missing values in "Total Charges" and "Tenure" were handled.
- ✓ **Integrity:** Data types were standardized for accurate financial calculations.

2. Strategic Business KPIs

26.54%

Churn Rate

7,043

Total Base

32.4m

Avg Tenure

\$139K

Monthly Loss

\$64.76

ARPU

932

VIP Churn

\$2,071

Estimated Avg Lifetime Value (CLV)

3. Deep Dive Insights

A. The Fiber Optic Anomaly

Our analysis revealed a significant switching rate in Fiber Optic technology (18.42%). Despite being a high-speed service, it has the highest churn compared to DSL, suggesting technical issues or pricing dissatisfaction.

B. Contractual Influence (CLT)

Customer Loyalty Tracking shows that "Month-to-month" contracts are the primary churn drivers. Long-term contracts (1-2 years) show a 3x higher retention rate.

C. Revenue Leakage

Churn is not just a volume issue; we are losing high-bill customers. This "High-Value Churn" significantly impacts the Monthly Recurring Revenue (MRR).

4. Data-Driven Recommendations

1. **Technical Audit:** Immediate review of Fiber Optic service quality and customer support response times.
2. **Migration Strategy:** Launch targeted campaigns to transition "Month-to-month" users into long-term contracts via loyalty discounts.
3. **VIP Retention:** Proactive outreach and dedicated account management for customers with bills above average to protect high-revenue streams.